

Ritvik Sharma

New York City, NY

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Education

Columbia University

Aug 2025 - Dec 2026

MS in Computer Science - AIML & NLP Track

SRM University – Rank: 2/700+

Jun 2019 - May 2023

BTech in Computer Science - AI & ML Track

GPA: 9.86/10

Work Experience

Barclays, UK

Jul 2023 - Jul 2025

Software Engineer-II

Pune, India

- **Saved \$200,000 Annually:** Innovated utilities ensuring optimal AWS Lambda Usage. **Collaborated** cross-functionally to drive adoption. Awarded a **Barclays-Wide Recognition**.
- **Achieved 96% Performance Improvement** in Glue Jobs pertaining to CRUD operations in AWS DynamoDB dealing with 5M+ Records (**12 hours to 27 minutes**). **Led** the solution design through Spark, Dynamic Frames, and Batch Write, **coordinating with platform teams** for seamless implementation.
- **Reduced error-resolution times by 75%** by designing and developing a complex error-routing utility via Python-based Lambdas, IBM Netcool, and SNS filters. Played a **key role in facilitating communication** between development and operations teams to align on system-wide logging strategies.
- **Executed No-Downtime Deployments:** Integrated Blue-Green and Canary strategies into the system, **advised leadership, aligned cross-domain teams**, and ensured uninterrupted user experience—earning Barclays-wide recognition.
- **Optimized CI-CD Pipelines with 63% higher success rates:** Fostered team collaboration across DevOps and application teams, earning organization-wide recognition at Barclays.
- **Spearheaded the engineering and development** of the integration layer connecting Onboarding, Rewards, Payments, and other modules for Barclays' first cloud-based credit card platform, **ArrowCard**.

Projects

VELVÉT - Luxury E-Commerce Website

[Link to Live Demo](#)

- Built an end-to-end luxury e-commerce site with React, Gatsby, AWS Lambda, Route53, CloudFront, DynamoDB.
- Synthesized ML-based semantic search by training a Custom BERT Model and hosting on AWS SageMaker. Leveraged AWS OpenSearch for fast product discovery.
- Implemented a secure and high quality AWS backend via IAM Roles, Basic DDoS Prevention and performed Selenium-based unit + integration tests to ensure Production-Grade Quality.

Inter-University Hackathon Winner: Mercury

[Link to Web/Video Demo](#)

- **Developed RESTful APIs** for an AI-powered chatbot leveraging ML and NLP to accurately extract food order details from natural language inputs such as, "The people want **2 Veg Margherita Pizzas** with **Olives** from **Dominos**".
- **Directed a team of 3 to architect and deliver** the backend of a mobile app using Python and Flutter, integrating ALBERT (LLM for NLP), Conditional Random Fields for probabilistic prediction, and Speech-to-Text capabilities.
- **Collaborated & orchestrated team victory** in a 24-hour hackathon by strategically allocating tasks based on **analyzing individual strengths**, demonstrating exceptional **teamwork and impactful communication**.

Technical Skills

Languages: Python, C/C++, JavaScript, SQL

Databases: MySQL, PostgreSQL, AWS DynamoDB, MongoDB, Oracle SQL

Developer Tools: Git, GitHub, Jenkins, Docker, Kubernetes, Kafka, Linux

Technologies: Amazon Web Services (AWS), Tomcat, Django, NodeJS, Selenium, gRPC, RESTful APIs, REST API, PyTorch, NLP, Pandas, NumPy, Scikit-Learn, BERT, Machine Learning, Computer Vision, OpenCV

Publications & Awards

- Sharma, Ritvik (April 2025). "VectorQuest: Faster and Better Open-Domain Question-Answering System" **Published in Springer Nature Switzerland AG** for the innovation leading to **50 times Faster Prediction**.
- Inter-University Hackathon Winner for Building an AI-Powered Chatbot for ordering food. [Check Out the Demo Here.](#)